From Tangent Lines to Convex Duality and the Geometry of L^p

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A derivative is a local linear approximation. For a differentiable function f, the tangent approximation at a is

$$f(a+h) = f(a) + f'(a)h + o(|h|).$$

When f is convex, that local line provides a *global* lower bound. This observation is the route from elementary calculus to Young's inequality, then to Holder's and Minkowski's inequalities, and finally to the duality between L^p norms. Our convex-analytic conventions follow Boyd and Vandenberghe, *Convex Optimization*, Section 3.3, together with the L^p material in MIT 18.102.[1]

§1.1 What Tangent Lines Remember

The derivative begins as a limit of secant slopes,

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h},$$

and differentiability means

$$f(a+h) = f(a) + f'(a)h + r(h), \qquad \lim_{h\to 0} \frac{r(h)}{|h|} = 0.$$

The linear term is the best first-order local description of the function. For a twice differentiable function, the next term is

$$f(a+h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + o(h^2).$$

This expansion reveals why the usual first-derivative test looks for changes in the sign of f' and the second-derivative test looks for the curvature f'': both are ways of detecting what first survives when the leading term disappears.

One should not silently identify a stationary point with an extremum. The function $f(x) = x^3$ has $f'(0) = 0$, but takes both signs on either side of zero. Meanwhile $f(x) = |x|$ has a strict minimum at zero *without* having a derivative there. Fermat's theorem says that differentiability plus an *interior extremum* implies $f' = 0$; the converse need not hold, and nondifferentiable extrema are not excluded.

Convexity adds something much stronger. For a differentiable convex function and $0 < t < 1$,

$$f(x + t(y-x)) \le (1-t)f(x) + tf(y).$$

Rearranging, dividing by $t > 0$, and letting $t \downarrow 0$ gives

$$f'(x)(y-x) \le f(y) - f(x).$$

So the tangent line at x is a *global* supporting line. This is exactly the inequality used below to derive Young's inequality. Convexity converts local slope data into an everywhere-valid comparison.

[1] Boyd and Vandenberghe, *Convex Optimization*, Chapter 3; MIT 18.102, functional-analysis lecture notes.

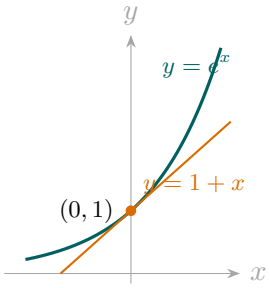


Figure 1.1: The exponential function lies globally above its tangent at zero: $e^x \geq 1 + x$. This one geometric fact is a prototype for the convex-analytic inequalities that follow.

§1.2 Convexity turns a tangent into an inequality

A differentiable function f on an interval is convex if and only if its graph lies above every tangent:

$$\boxed{f(y) \geq f(x) + f'(x)(y - x).}$$

For a twice differentiable function, $f'' \geq 0$ guarantees this property. Indeed, by the mean value theorem applied to f', its derivative is nondecreasing, so the secant slopes to the right of x are no smaller than $f'(x)$; the same estimate follows on the left.

The simple convex function $f(t) = e^t$ satisfies $e^t \geq 1 + t$ by taking the tangent at 0. For $f(t) = -\log t$ on $t > 0$, the tangent at 1 gives

$$-\log t \geq 1 - t, \qquad \text{or} \qquad \log t \leq t - 1.$$

These are global inequalities obtained from a first-order local calculation. Equality holds at the tangency point.

§1.3 Young's inequality is a conjugate-function calculation

Let $p > 1$ and $q = p/(p-1)$. Consider the convex function

$$\phi(u) = \frac{|u|^p}{p}.$$

For each real v, its convex conjugate is

$$\phi^*(v) = \sup_{u \in \mathbb{R}} \{uv - \phi(u)\}.$$

The optimizing u has the sign of v, and the stationarity equation is

$$|u|^{p-1} = |v|.$$

Thus $u = \operatorname{sgn}(v)|v|^{q-1}$, and substitution gives

$$\boxed{\phi^*(v) = \frac{|v|^q}{q}.}$$

By the definition of a supremum, $uv - \phi(u) \leq \phi^*(v)$. We obtain Young's inequality

$$\boxed{uv \leq \frac{|u|^p}{p} + \frac{|v|^q}{q}.}$$

Equality holds exactly when $v = \operatorname{sgn}(u)|u|^{p-1}$.

This explains the conjugate exponents more sharply than memorizing $1/p + 1/q = 1$. The power q appears because the inverse of $u \mapsto |u|^{p-1}\operatorname{sgn}(u)$ is $v \mapsto |v|^{q-1}\operatorname{sgn}(v)$.

§1.4 Holder's inequality is Young after normalization

For nonnegative numbers a_i, b_i, put

$$A = \left(\sum_i a_i^p\right)^{1/p}, \qquad B = \left(\sum_i b_i^q\right)^{1/q}.$$

If $A = 0$ or $B = 0$, the inequality below is immediate. Otherwise Young's inequality applied to a_i/A and b_i/B gives

$$\frac{a_i b_i}{AB} \le \frac{a_i^p}{pA^p} + \frac{b_i^q}{qB^q}.$$

Sum over i. The right-hand side becomes $1/p + 1/q = 1$, yielding

$$\boxed{\sum_i a_i b_i \le \left(\sum_i a_i^p\right)^{1/p} \left(\sum_i b_i^q\right)^{1/q}.}$$

For arbitrary signed or complex sequences, replace each product by its absolute value. Exactly the same pointwise argument, with integration in place of summation, proves

$$\int |fg| \le \|f\|_{L^p}\|g\|_{L^q}$$

whenever the respective norms are finite.

There is a revealing dual formulation in finite-dimensional real spaces (for complex vectors, use the absolute value of the Hermitian pairing):

$$\boxed{\|x\|_p = \sup_{\|y\|_q \le 1} \sum_i x_i y_i.}$$

Holder gives the upper bound. If $x \neq 0$, choose

$$y_i = \frac{\operatorname{sgn}(x_i)|x_i|^{p-1}}{\|x\|_p^{p-1}}.$$

Then $\|y\|_q = 1$, and $\sum_i x_i y_i = \|x\|_p$. The dual vector is exactly the equality case of Young's inequality, now interpreted as the linear measurement achieving the norm.

§1.5 Minkowski makes ℓ^p into a normed space

For $p > 1$, let $z = x + y$. By Holder with exponents p, q,

$$\begin{aligned}
\|z\|_p^p &= \sum_i |z_i|^p \\
&\le \sum_i |x_i||z_i|^{p-1} + \sum_i |y_i||z_i|^{p-1} \\
&\le (\|x\|_p + \|y\|_p)\left(\sum_i |z_i|^{(p-1)q}\right)^{1/q} \\
&= (\|x\|_p + \|y\|_p)\|z\|_p^{p-1}.
\end{aligned}$$

If $z = 0$, the conclusion is trivial; otherwise divide by $\|z\|_p^{p-1}$ to obtain

$$\boxed{\|x + y\|_p \leq \|x\|_p + \|y\|_p.}$$

The same reasoning applies to L^p integrals, while $p = 1$ follows directly from the pointwise triangle inequality.

There is a conceptual difference between proving an inequality and proving a norm axiom. Holder controls the pairing of vectors from conjugate exponent spaces; Minkowski uses that control to show that ℓ^p is closed under addition with a subadditive notion of size. The norm's geometry is therefore built from its duality inequality.

§1.6 Why the exponent threshold really matters

Young's inequality and its convex-conjugate proof require $p > 1$, or $p = 1$ with a limiting dual exponent $q = \infty$. What happens if we blindly extend the expression

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{1/p}$$

to $0 < p < 1$? The expression is homogeneous and still vanishes only at zero, but it can fail the triangle inequality.

For example, take $p = 1/2$ and

$$x = (1, 0), \qquad y = (0, 1).$$

Then

$$\|x\|_{1/2} = \|y\|_{1/2} = 1, \qquad \boxed{\|x + y\|_{1/2} = (1 + 1)^2 = 4 > 2.}$$

The unit ball for this *quasi-norm* is not convex. The proof of Minkowski breaks at precisely the same place: one would need the conjugate exponent $q = p/(p-1) < 0$, and the convexity argument for the power function no longer applies.

The equality case of Holder is also geometric. When $p = q = 2$, it reduces to Cauchy–Schwarz; equality for nonzero real vectors means that the vectors are proportional. The resulting Minkowski triangle inequality is strict for two nonzero vectors pointing in genuinely different directions. For $x = (1, 0)$ and $y = (0, 1)$,

$$\|x + y\|_2 = \sqrt{2} < 2 = \|x\|_2 + \|y\|_2,$$

whereas if $y = tx$ with $t \geq 0$ the triangle inequality becomes equality. This is why p-norms are not merely formulas for adding powers: their convexity properties determine the geometry of the unit ball and which inequalities may be used.

§1.7 Jensen's inequality and entropy

For a convex function f, nonnegative weights α_i with $\sum_i \alpha_i = 1$ satisfy

$$f\left(\sum_i \alpha_i x_i \right) \leq \sum_i \alpha_i f(x_i).$$

This is Jensen's inequality, obtained by repeatedly applying the two-point definition of convexity. The direction reverses for concave functions such as log.

A particularly useful application involves probability vectors $p = (p_i)$, $q = (q_i)$ with strictly positive coordinates and total mass 1. Concavity of log gives

$$\sum_i p_i \log \frac{q_i}{p_i} \le \log\left(\sum_i p_i \frac{q_i}{p_i}\right) = \log \sum_i q_i = 0.$$

Therefore the relative entropy, or Kullback–Leibler divergence, satisfies

$$\boxed{D_{\mathrm{KL}}(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} \ge 0.}$$

Equality, by strict concavity, holds if and only if $p = q$. The result extends to zeros by the standard conventions and appropriate support conditions.

Taking q uniform on N points gives

$$H(p) = -\sum_i p_i \log p_i \le \log N,$$

with equality at the uniform distribution. If there are 2^n states and base-two logarithms are used, this reads $H(p) \le n$.

The connection to entropy emerges from precisely the same tangent and convexity technology that produced Young's inequality. Convexity controls a function by linear data; conjugacy controls a bilinear pairing; Jensen controls averages. These three operations provide a common analytic language for norms, optimization, and information.